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CANARY: Collision-Free Audio Watermarking for Proactive Deepfake Detection

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Abstract

The rapid proliferation of generative audio models poses severe security risks to digital identity and content copyright. This paper presents CANARY, a proactive deepfake detection framework based on collision-free acoustic watermarking. By embedding pseudo-random orthogonal sequence codes in the high-frequency Discrete Cosine Transform (DCT) domain of audio signals prior to dissemination, CANARY ensures imperceptibility and prevents watermark collisions when multiple audio streams are merged or re-encoded.

Key Methodology & Architecture

DCT-based orthogonal sequence embedding, deep neural decoder network, perceptual acoustic masking (ITU-R BS.1387), collision-avoidance hash indexing.

Datasets & Benchmark Environments

LibriSpeech, ASVspoof 2021, VoxCeleb2 (Over 10,000 speech samples).

Performance Metrics & Graph Axis Parameters

Reported Results: Bit Error Rate (BER) < 0.05%, Watermark Detection Accuracy = 98.7%, PSNR = 46.2 dB.

Graph Parameter Mapping: X-axis: Signal-to-Noise Ratio (SNR in dB, 0 to 40 dB); Y-axis: Bit Error Rate (BER, 0.0 to 0.5).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Proactive protection, collision-free multi-stream tracking, robust against neural vocoders.	Slightly higher computational latency during initial watermark embedding phase.

IEEE Citation Format

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